

Customer Segmentation & Churn Pattern Analytics in European Banking

Business Analysis & Management Report

Prepared for Management Review

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1. Executive Summary

The analysis was undertaken to understand how customer churn is distributed across the European banking customer base and to identify the customer groups that require the closest retention attention. The work used 10,000 customer records covering France, Germany and Spain and examined geography, age, credit score, tenure, balance, gender, product usage and customer activity.

20.37%

Overall churn rate

2,037 of 10,000 customers are classified as churned

32.44%

Germany churn rate

Geographic Risk Index: 1.59

51.12%

46-60 churn rate

Highest standalone segment churn

24.16%

High-value churn rate

High-balance customers used as a value proxy

12.58 pp

Engagement gap

Inactive 26.85% vs active 14.27%

24.18%

High-value balance exposure

110.80 million associated with churned high-value customers

Management takeaway

The results point to a concentrated retention problem rather than a uniform churn problem. Germany, customers aged 46-60, inactive customers and high-balance customers are the main areas of concern. The strongest combined pattern is Germany + age 46-60, where churn reaches 67.33%. Within the high-value group, Germany + age 46-60 reaches 65.06%.

2. Business Problem and Objectives

The project addresses a practical gap in churn management: an overall churn rate does not show which customer groups are most exposed or where retention effort should be concentrated.

Business questions addressed:

- Which customer groups have the highest observed churn rates?
- How does churn differ across France, Germany and Spain?
- Which age groups and customer lifecycle groups require closer attention?
- Is churn concentrated among customers with larger balances?
- Does customer inactivity coincide with higher churn?
- Which segments contribute the largest number of churned customers?

Primary objectives included measuring overall churn, identifying segment-level churn patterns, comparing European regions, understanding high-value customer churn, evaluating engagement and tenure patterns, and supporting targeted retention planning.

3. Data and Analytical Approach

The source dataset contains 10,000 customer records and 14 fields. The analytical preparation removed CustomerId, Surname and the constant Year field, leaving 11 analytical variables. Segmentation fields were then added for Geography, Age, Credit Score, Tenure and Balance.

Area	Approach
Geography	France, Germany, Spain
Age	<30, 30-45, 46-60, 60+
Credit Score	Low, Medium, High
Tenure	New, Mid-term, Long-term
Balance	Zero-balance, Low-balance, High-balance
High-value proxy	High-balance customers

The analysis was descriptive and segmentation-based. Churn rate measures the share of customers who churned within a segment; churn contribution measures the segment share of all churners. Both were used because a high-rate segment and a high-volume segment can be different.

4. Customer Base and Data Quality

The initial data validation found 10,000 rows and 14 source columns. All fields were fully populated, with no missing values and no duplicate rows. CustomerId was unique for all 10,000 customers. Geography contained three expected countries and the binary fields HasCrCard, IsActiveMember and Exited contained consistent 0/1 values.

Measure	Result
Records	10,000
Source columns	14
Analytical columns after cleaning	11
Missing values	0
Duplicate rows	0
Churned customers	2,037
Retained customers	7,963
Overall churn	20.37%

The Year field contained only 2025 and therefore had no analytical variation. High balances and high salaries were retained rather than treated as automatic errors because financially significant customers are central to the project.

5. Segmentation Findings

The customer base is reasonably distributed across the required analytical segments, with two notable concentration patterns: France represents half of the customer base, and the 30-45 age group represents 62.48%.

Segment dimension	Key distribution
Geography	France 50.14%; Germany 25.09%; Spain 24.77%
Age	<30 16.41%; 30-45 62.48%; 46-60 16.47%; 60+ 4.64%
Credit score	Low 30.34%; Medium 38.18%; High 31.48%
Tenure	New 35.05%; Mid-term 29.68%; Long-term 35.27%
Balance	Zero 36.17%; Low 31.92%; High 31.91%

The balance distribution is particularly relevant. 36.17% of customers have a zero balance, so zero-balance customers were kept as a separate group rather than mixed with low positive balances. High- and low-balance segments were split using the median positive balance of 119,839.69.

6. Churn Pattern Findings

The overall churn rate is 20.37%, but the segment results are not evenly distributed.

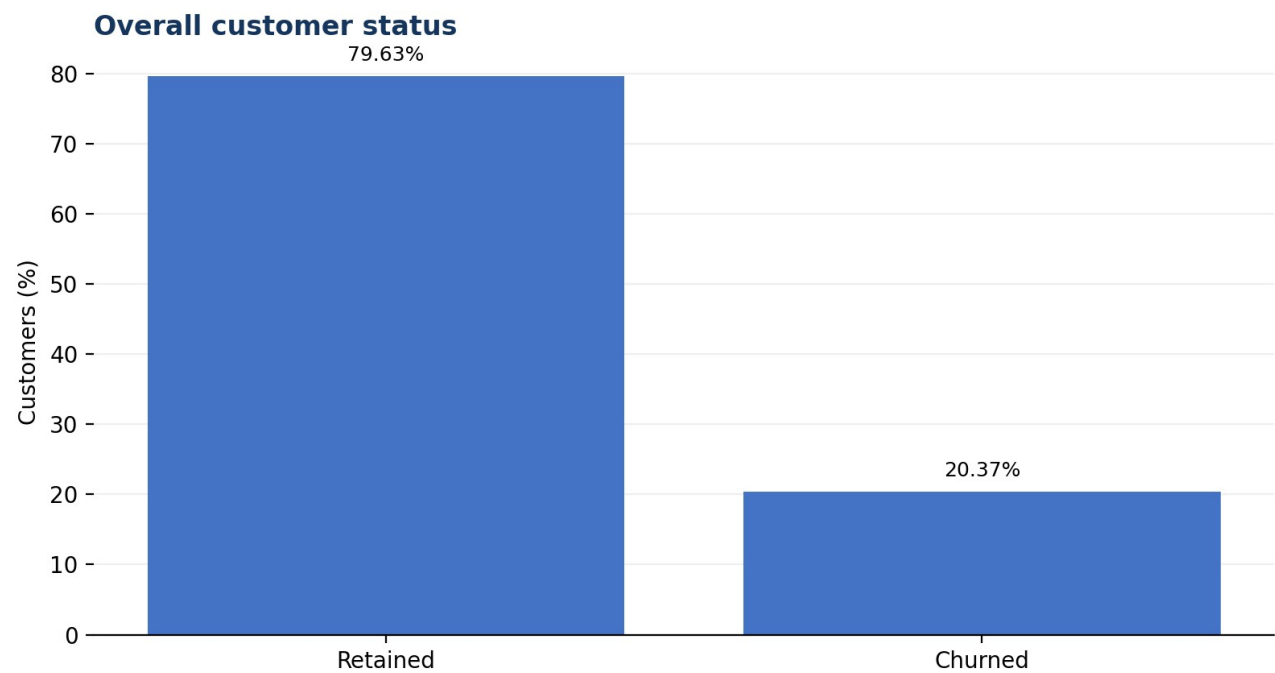


Figure 1. Overall retained versus churned customer distribution.

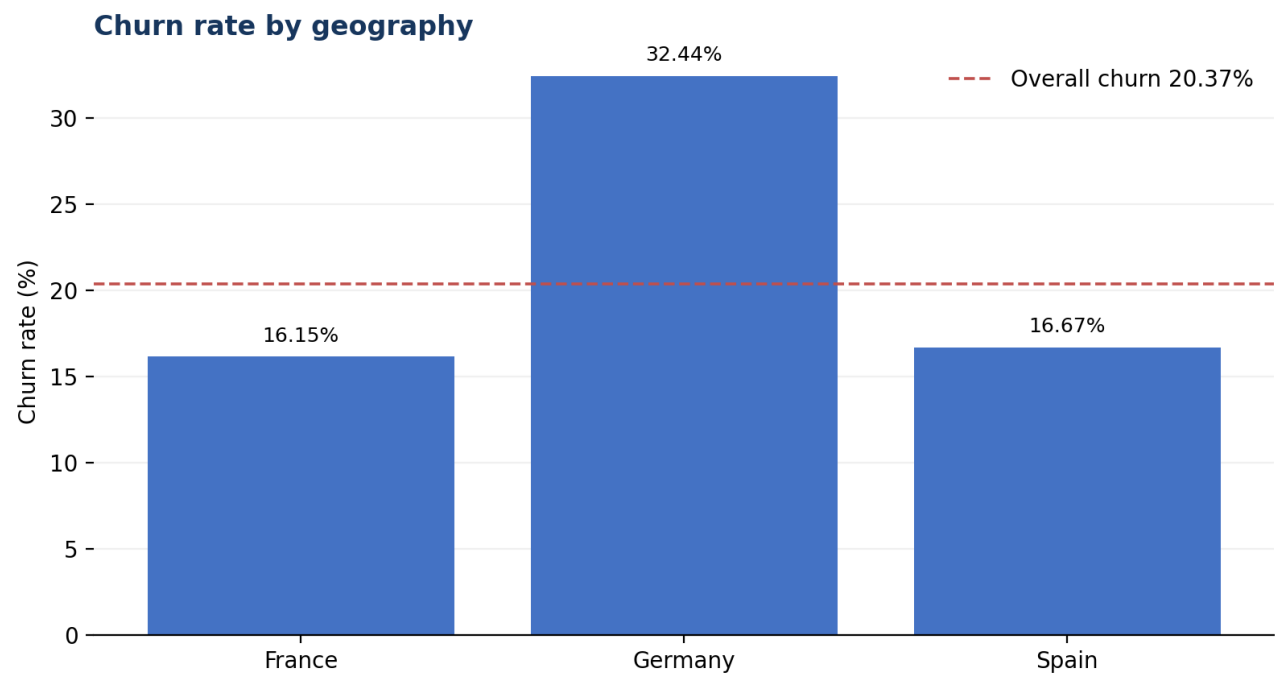


Figure 2. Germany is materially above the overall churn baseline.

Dimension	Lowest churn	Highest churn	Main observation
Geography	France 16.15%	Germany 32.44%	Germany has the highest geographic churn
Age	<30 7.56%	46-60 51.12%	Age shows the strongest standalone separation
Credit score	Medium 19.72%	Low 21.75%	Differences are relatively small
Tenure	Long-term 19.51%	New 21.14%	Tenure shows a mild

			difference
Balance	Zero 13.82%	High 24.16%	Positive-balance customers show higher churn

The 30-45 group has the largest contribution to total churn at 46.93% because it is the largest customer group, even though its own churn rate is 15.30%. This distinction between churn rate and churn contribution is important for management prioritisation.

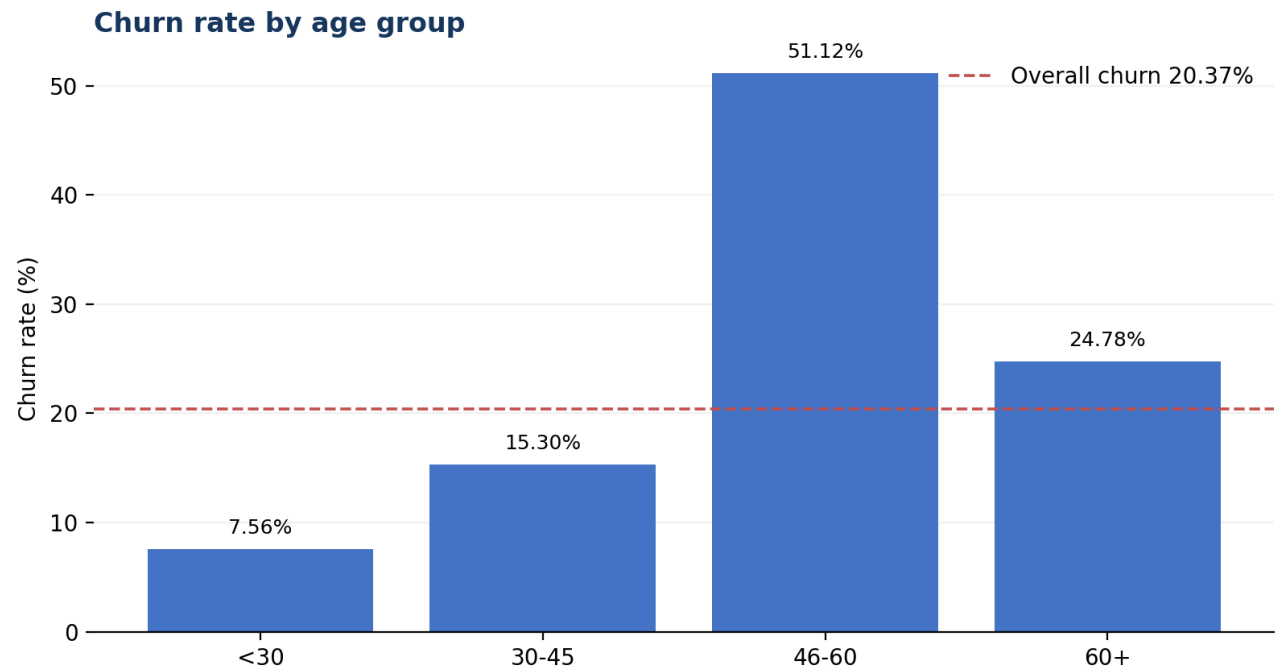


Figure 3. Age is the strongest standalone segmentation signal in the current dataset.

7. Comparative Demographic Analysis

Gender, geography-age interactions and financial characteristics were compared to identify patterns that are not visible in one-dimensional segment views.

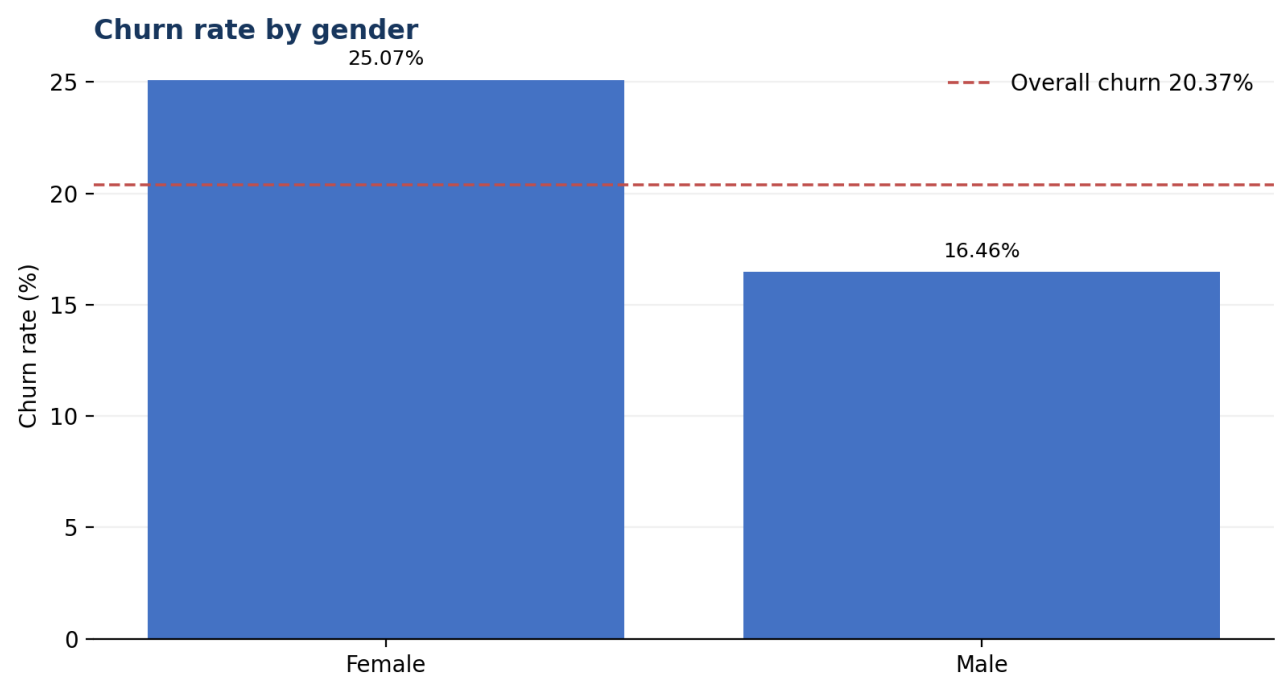


Figure 4. Female customers have a higher observed churn rate than male customers.

Gender	Customers	Churned	Churn rate	Churn contribution
Female	4,543	1,139	25.07%	55.92%
Male	5,457	898	16.46%	44.08%

Female customers have an observed churn rate 8.61 percentage points above males and contribute 55.92% of total churn despite being the smaller population. This should be treated as a segment-level association and investigated alongside geography, age and engagement before any targeted policy is introduced.

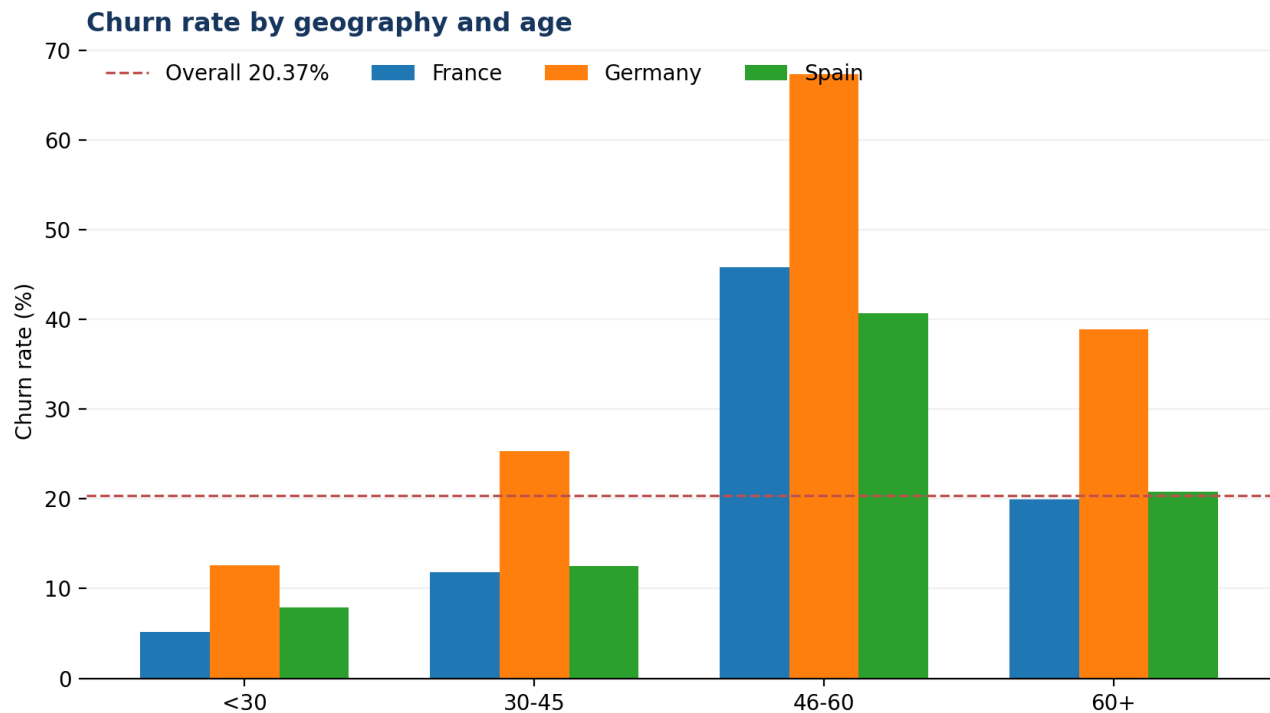


Figure 5. Geography x age interaction shows that the 46-60 pattern persists across all countries.

Country-age group	Customers	Churners	Churn rate
Germany + 46-60	502	338	67.33%
France + 46-60	749	343	45.79%
Spain + 46-60	396	161	40.66%
Germany + 30-45	1,522	385	25.30%

The strongest interaction is Germany + 46-60, where 338 of 502 customers churned. The pattern is not isolated to Germany, however: the 46-60 group also has elevated churn in France and Spain. This makes age a major area for further investigation.

8. High-Value Customer Analysis

High-value customers are defined in this report as high-balance customers. This is a proxy for financial significance, not a direct measure of customer profitability, revenue or lifetime value.

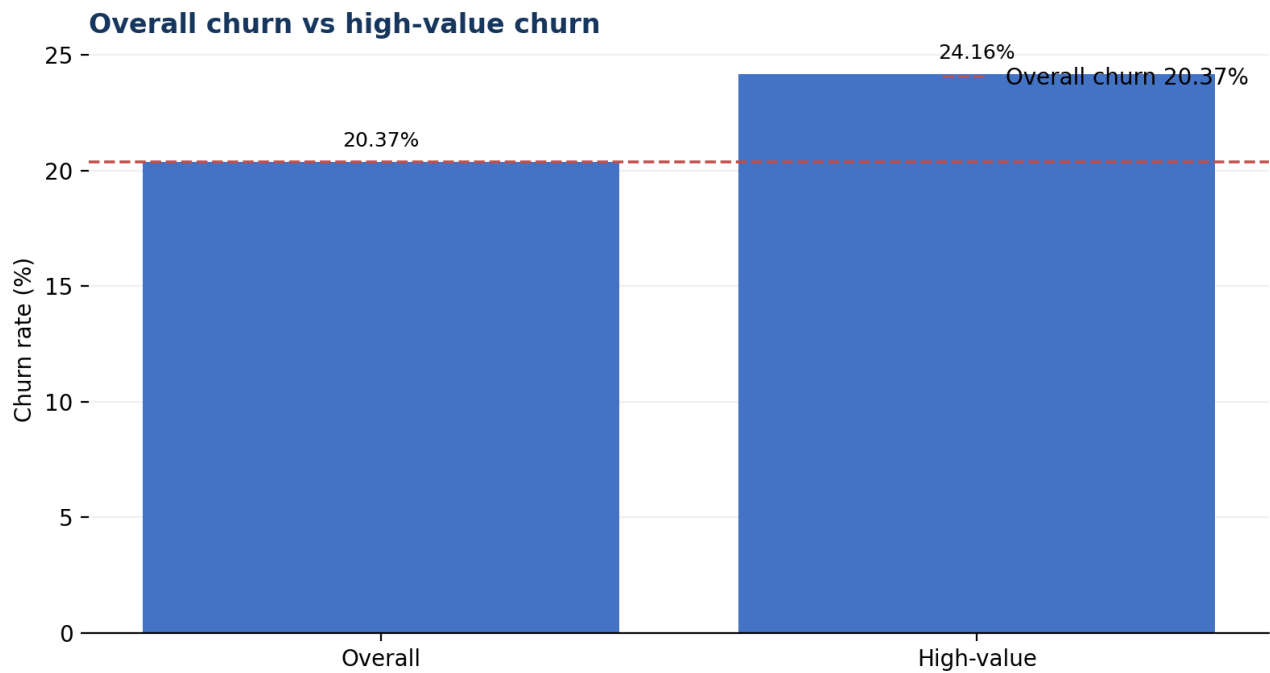


Figure 6. High-value customers have above-baseline churn.

Measure	High-value	Non-high-value
Customers	3,191	6,809
Share of customers	31.91%	68.09%
Churners	771	1,266
Churn rate	24.16%	18.59%
Share of all churners	37.85%	62.15%

High-value customers churn at 24.16%, 3.79 percentage points above the overall rate. They account for 37.85% of all churners, making them a meaningful retention priority.

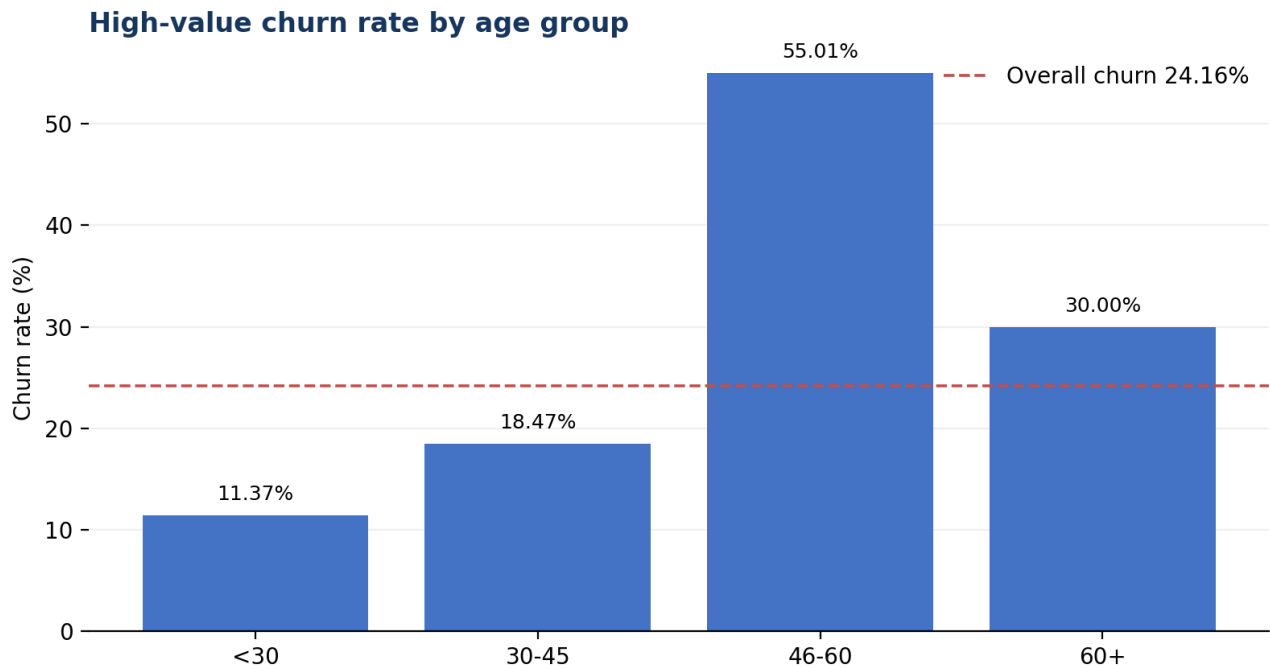


Figure 7. High-value churn is highest in the 46-60 age group.

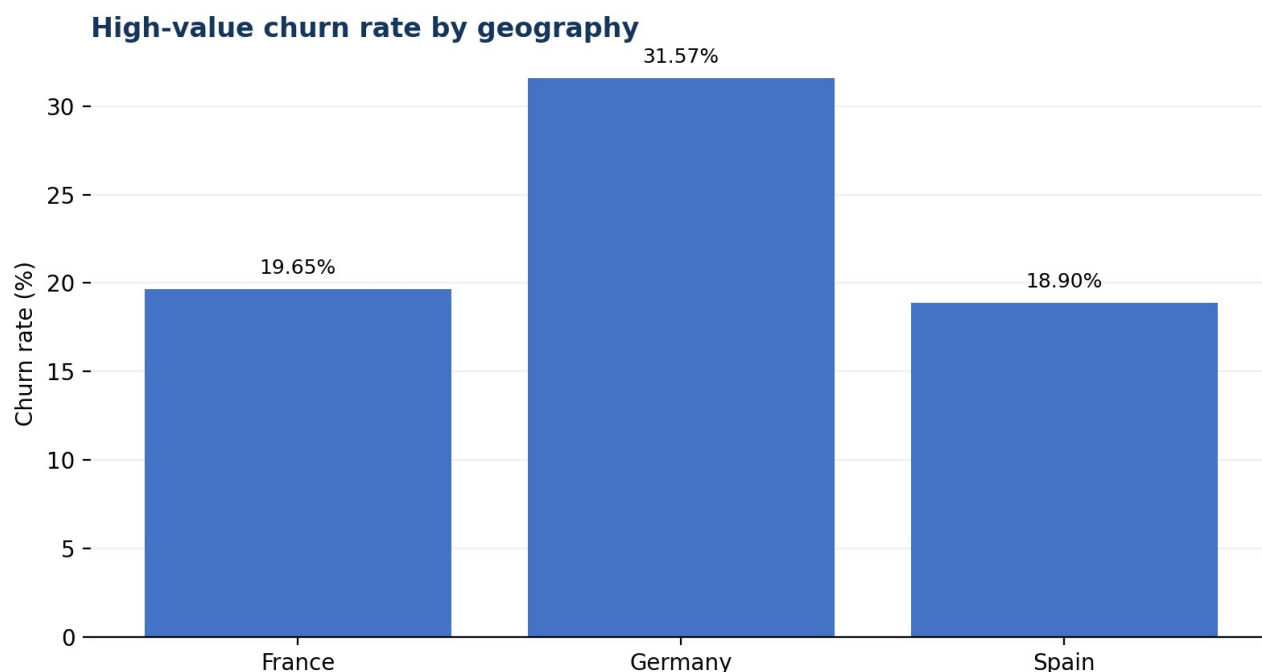


Figure 8. Germany has the highest high-value churn rate.

Within the high-value segment, Germany records a 31.57% churn rate and 51.10% of high-value churners. The 46-60 high-value group has a 55.01% churn rate. The most pronounced combined subgroup is Germany + 46-60 + high-value at 65.06%.

Financial exposure

Measure	Result
Total high-value customers	3,191
Total high-value balance	458.15 million
High-value churners	771
Balance associated with churned high-value customers	110.80 million
High-value balance exposure	24.18%

Approximately 24.18% of the high-value balance is associated with customers classified as churned. This is an exposure measure, not a confirmed revenue loss.

9. KPI Framework

KPI	Result	Interpretation
Overall Churn Rate	20.37%	Baseline for all segment comparisons
High-Value Churn Ratio	24.16%	Churn among high-balance customers
Germany Geographic Risk Index	1.59	Germany churn is above the overall baseline
Engagement Drop Indicator	12.58 pp	Inactive churn exceeds active churn by 12.58 points

High-Value Balance Exposure	24.18%	Share of high-value balance associated with churned customers
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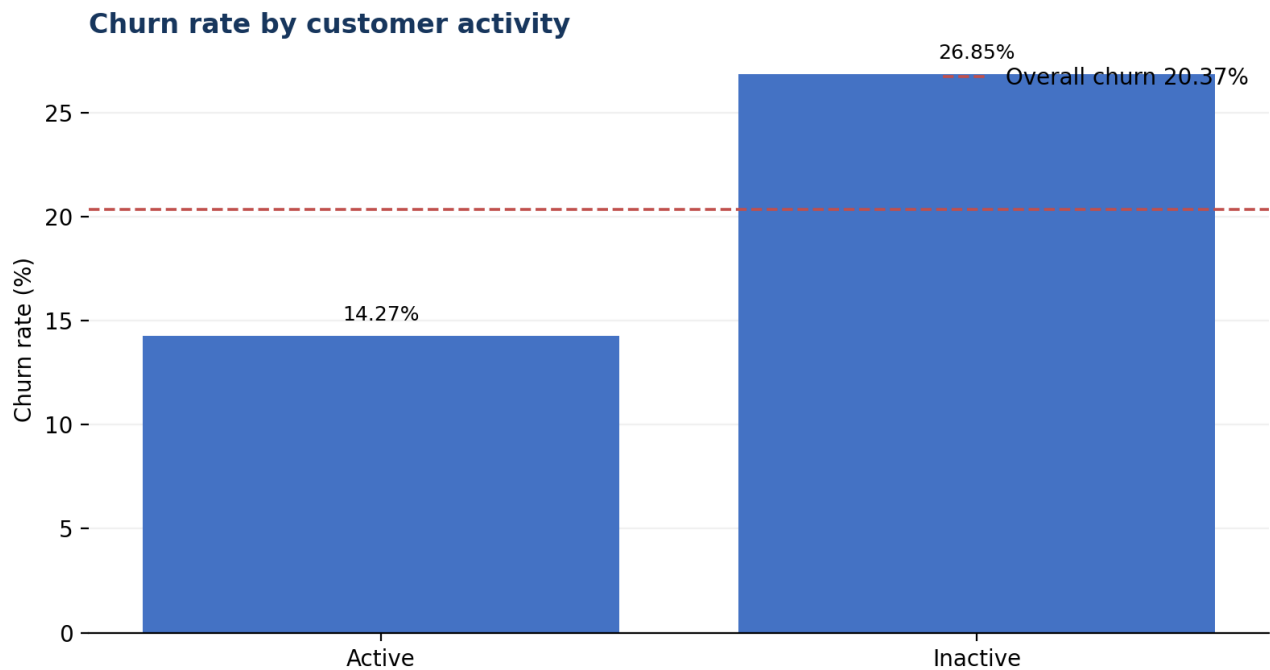


Figure 9. Inactive customers have materially higher churn than active customers.

The engagement analysis is one of the clearest results in the project. Active customers have 14.27% churn, while inactive customers have 26.85%, producing a 12.58 percentage-point gap. Among high-value customers, the gap widens to 13.91 percentage points (17.32% active vs 31.23% inactive).

10. Business Recommendations

The analysis supports a focused retention approach rather than a single intervention for the whole customer base.

- Prioritise Germany:** Use Germany as a priority monitoring geography, with deeper review of customer engagement, product usage and service experience.
- Focus on age 46-60:** Create a dedicated review of customers aged 46-60, particularly in Germany and among high-balance customers.
- Act on inactivity:** Introduce engagement monitoring and early-warning outreach for inactive customers, with tighter controls for high-value inactive accounts.
- Protect high-balance relationships:** Track high-balance churn and the balance exposure linked to churned accounts. Use this as a financial exposure KPI rather than a profitability measure.
- Use combined segments:** Operational reporting should allow drill-down by geography, age, activity and high-value status rather than relying only on overall churn.

Validate before targeting: Before customer-level intervention, validate these patterns using operational CRM, product, service and transaction data that are not present in the current dataset.

11. Limitations and Use of Findings

- The analysis is descriptive. It identifies associations in the dataset and does not establish cause and effect.
- The high-value definition is based on account balance because direct revenue, profit and customer lifetime value are not available.
- The dataset is cross-sectional and contains one year value (2025), so year-over-year churn trends cannot be assessed.
- Some small interaction groups have fewer customers and should be monitored with their population size alongside the churn rate.
- Operational recommendations should be validated with additional bank data before implementation.

12. Conclusion

The analysis shows that churn is concentrated in specific customer groups rather than evenly distributed across the banking population. Germany has the highest geographic churn, customers aged 46-60 show the strongest age-related churn, inactive customers have substantially higher churn than active customers, and high-balance customers represent meaningful financial exposure.

The strongest combined signal is the Germany + 46-60 segment, while the high-value analysis shows that this pattern also persists among customers with larger balances. The final KPI framework provides a practical way to monitor these patterns in management reporting and can be carried forward directly into the planned Streamlit dashboard.

The main management implication is clear: retention effort should be prioritised by segment, engagement and financial exposure, with regular monitoring rather than a one-size-fits-all churn program.

Appendix — Key Metrics at a Glance

Metric	Value
Customers analysed	10,000
Churned customers	2,037
Overall churn rate	20.37%
Germany churn rate	32.44%
46-60 churn rate	51.12%
Germany + 46-60 churn rate	67.33%
High-value customers	3,191
High-value churn rate	24.16%
High-value churners	771
Churned high-value balance	110.80 million
High-value balance exposure	24.18%
Inactive customer churn rate	26.85%
Active customer churn rate	14.27%
Engagement drop indicator	12.58 percentage points

Source: European_Bank.csv and completed project analysis outputs. Values are reported as observed in the project dataset.